

HTL-122-AT35MB FPGA Evaluation Board User's Guide

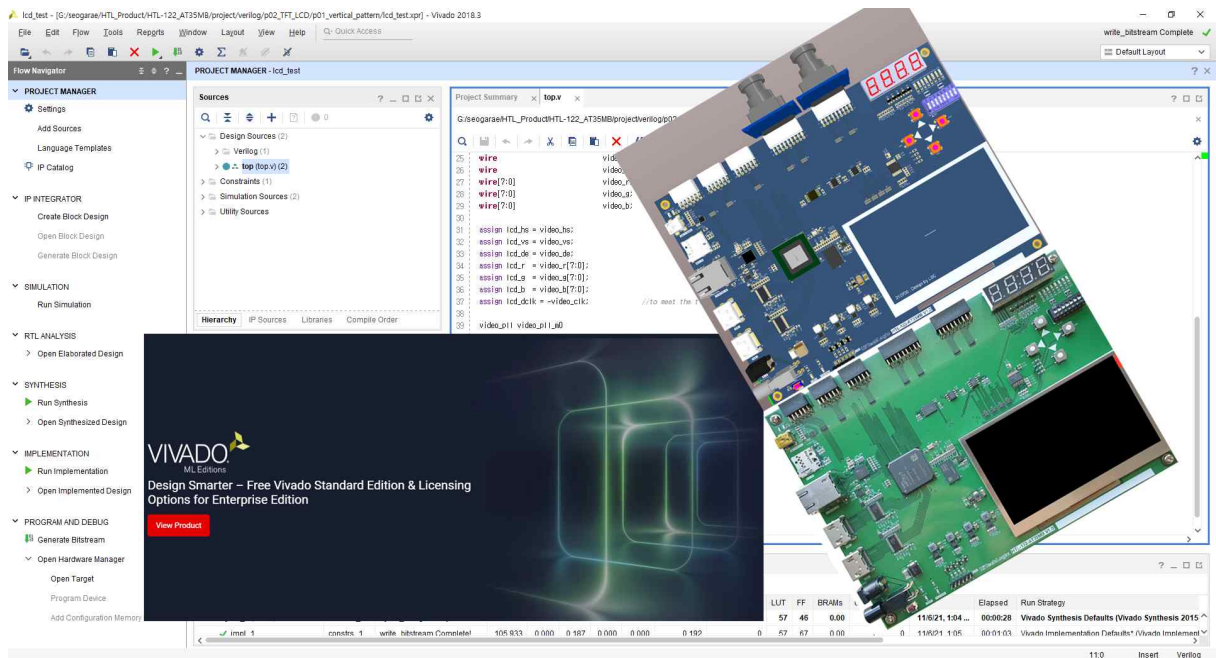


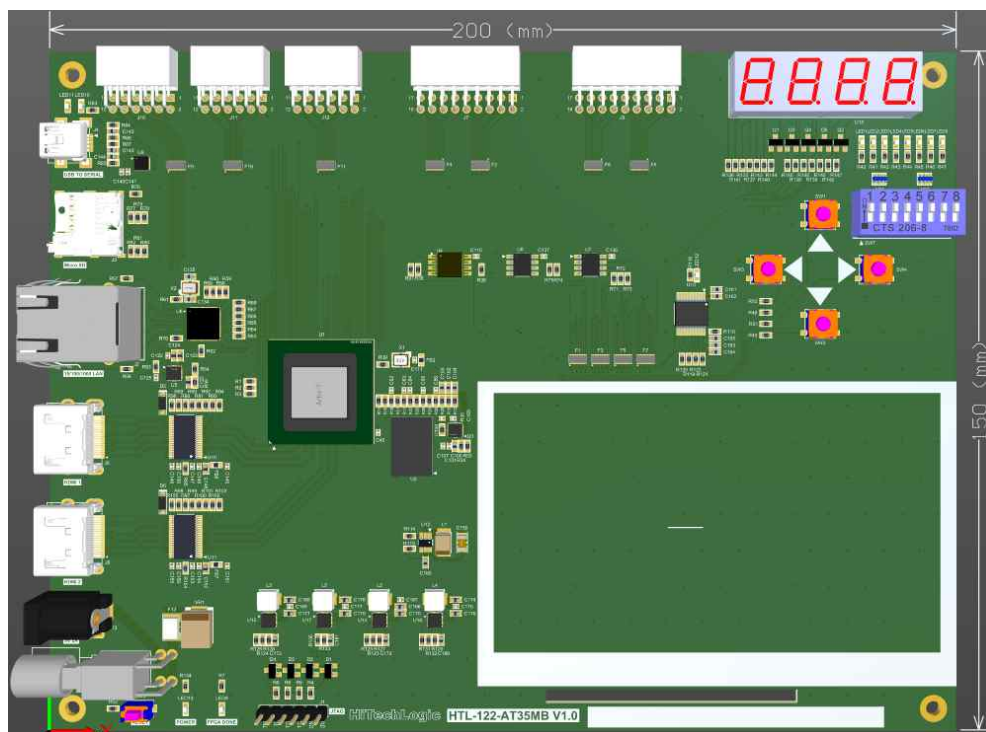
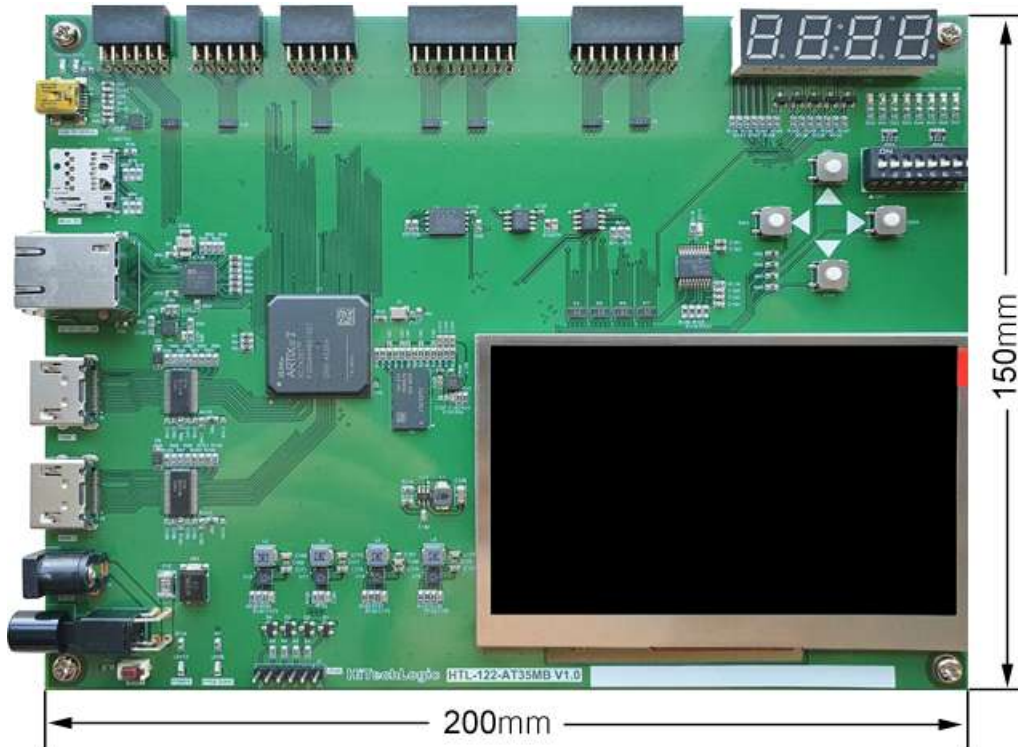
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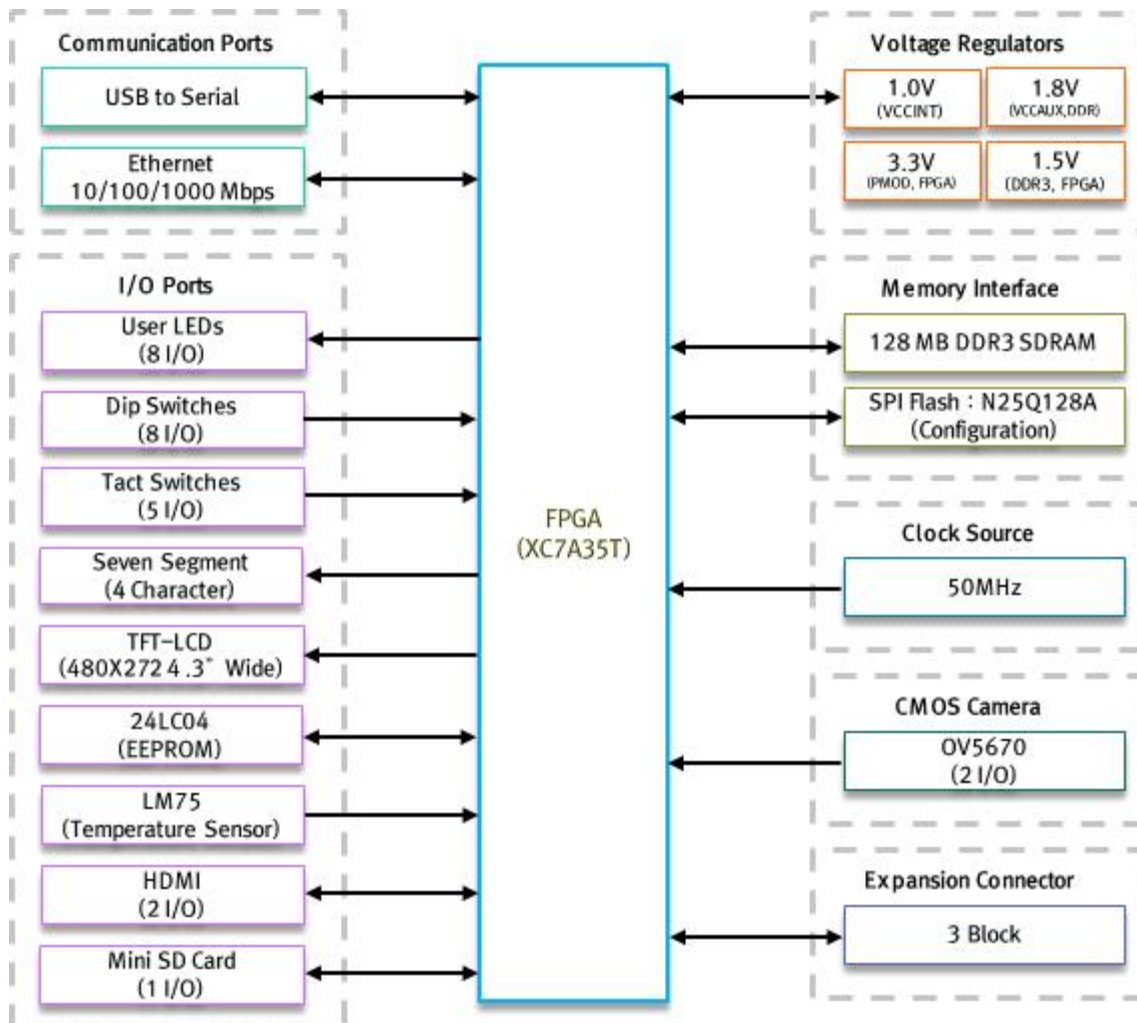
1. 제품 외형

1.1 기구도면

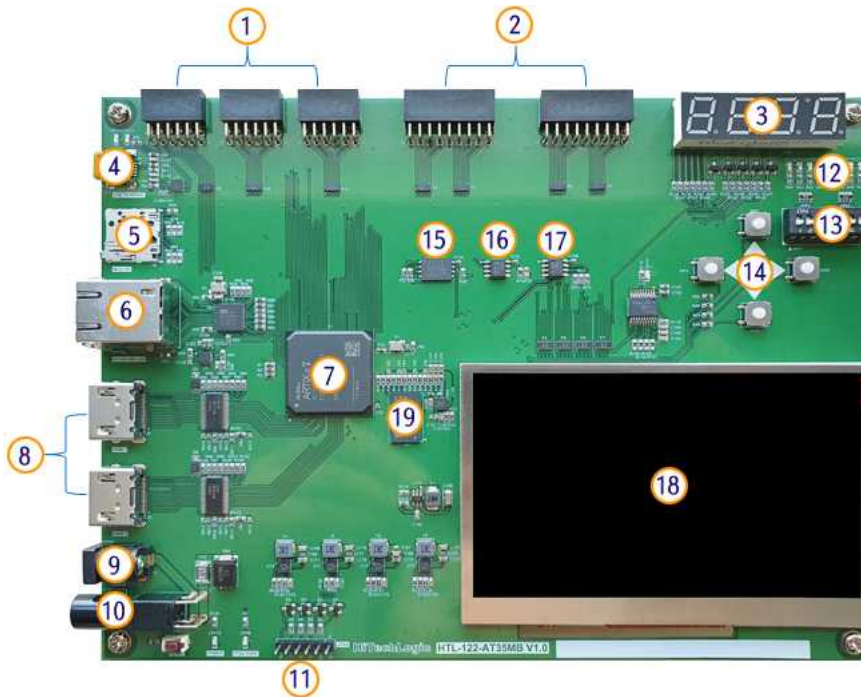
□ 단위 : mm



2. Function Description



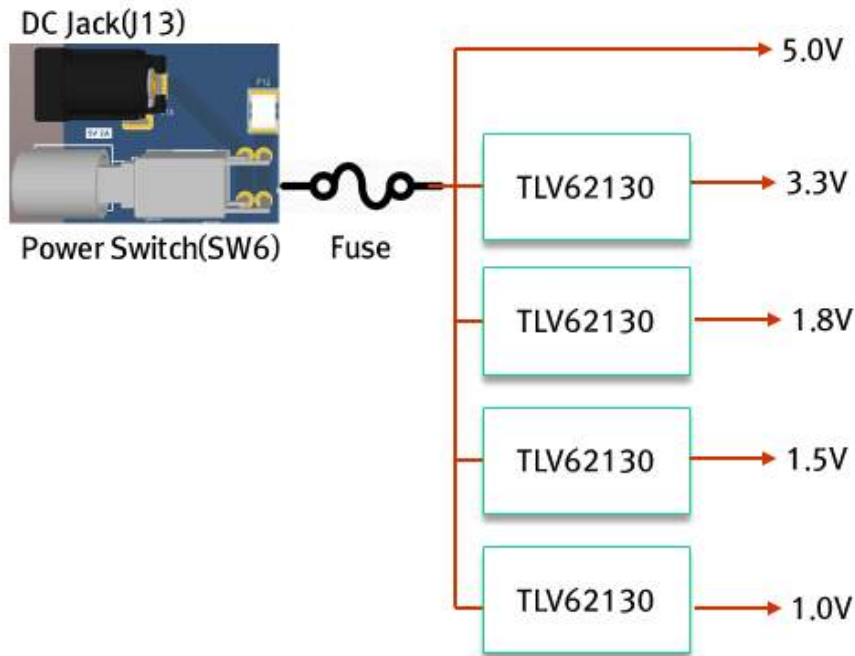
3. 보드 구성 및 명칭



- ① Expansion Connector : 3EA
- ② CMOS Image Sensor : OV5640(2EA)
- ③ Four Character Seven-Segment Digit
- ④ USB to Serial
- ⑤ Mini SD Card
- ⑥ Ethernet
- ⑦ FPGA : XC7A35T-FGG484
- ⑧ HDMI : 2EA
- ⑨ Power Input : Adapt(DC5V Input)
- ⑩ Power On/Off Switch
- ⑪ JTAG
- ⑫ 8 LED
- ⑬ 8 Input Dip Switches
- ⑭ 4 Push Switches
- ⑮ PROM : N25Q128
- ⑯ Temperature Sensor : LM75
- ⑰ EEPROM : 24LC04
- ⑱ TFT-LCD : 4.3Inch(480×272 Pixel)
- ⑲ DDR3

4. Power

HTL-123-AT35MB 보드의 모든 전원 공급 장치는 전원 스위치(SW6)로 ON/OFF 할 수 있으며, HTL-123-AT35MB 보드에서는 3.3V/1.8V/1.5V/1.0V의 전원을 사용한다.

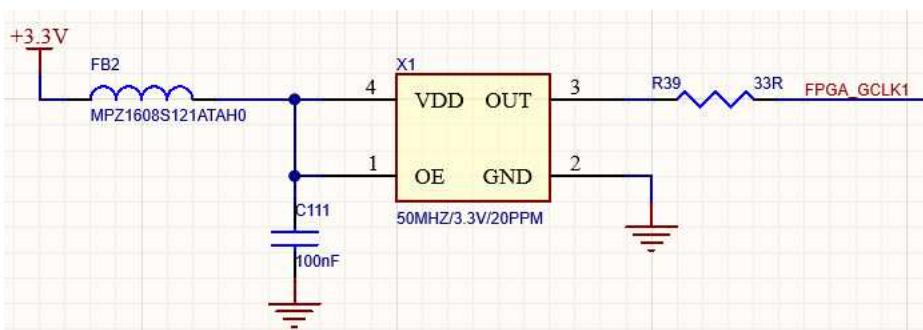


5. FPGA

HTL-123-AT35MB 보드에는 Artix-7 FPGA 패밀리 중 XC7A35T-2FGG484 탑재되어 있다.



6. Clock : 50MHz



7. JTAG

7.1 JTAG은 별도 구매이며, 함께 공급하지 않습니다. 아래 하이텍로직에서 생산하는 "Xilinx Platform Cable USB(XFC-XPCU)"를 디바이스마트에서 판매하는 URL 아래와 같다.

<https://www.devicemart.co.kr/goods/view?no=32498>

홈 > MCU보드/현자키트 > 프로세서/개발보드 > FPGA

[하이텍로직] Xilinx Platform Cable USB(XFC-XPCU)

Xilinx Platform Cable USB과 호환되는 제품으로서 Xilinx FPGA/CPLD 디바이스 지원 (5V 디바이스 제공되지 않음), ISE (IMPACT 지원)(ISE WebPACK 7.0 이상), USB 포트 제공

하이텍로직

상품번호 32498

판매가 (VAT 별도/포함) **90,000원** (99,000원)

사은품 매거진 사은품 (2021-07-01 ~ 2021-11-30) [?]

평균준비기간 2~3일

제조사 하이텍로직

수량 - 1 +

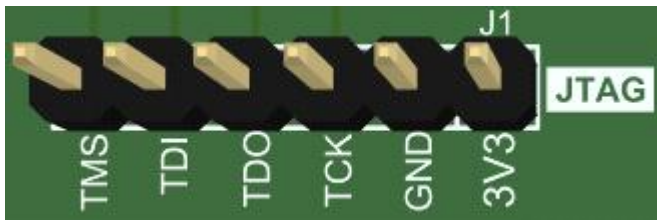
[주의] ★ 미사용이더라도 제품상자 개봉 시 반출 불가요니 반드시 개봉 전 참고 부탁드립니다.

총 상품금액 (VAT 별도) **90,000원**
(VAT 포함) 99,000원

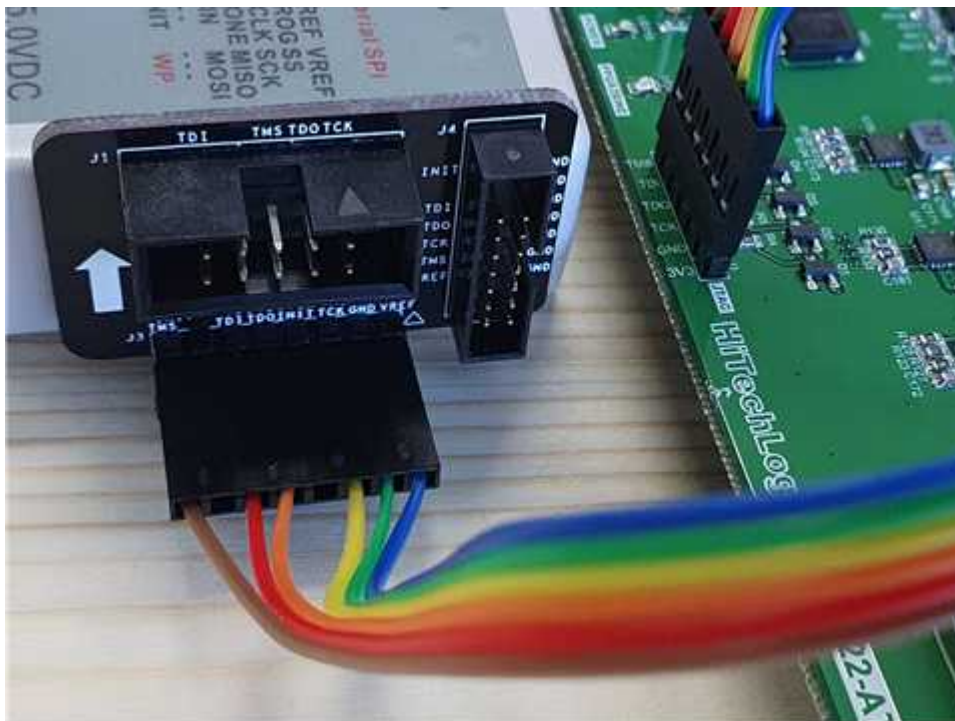
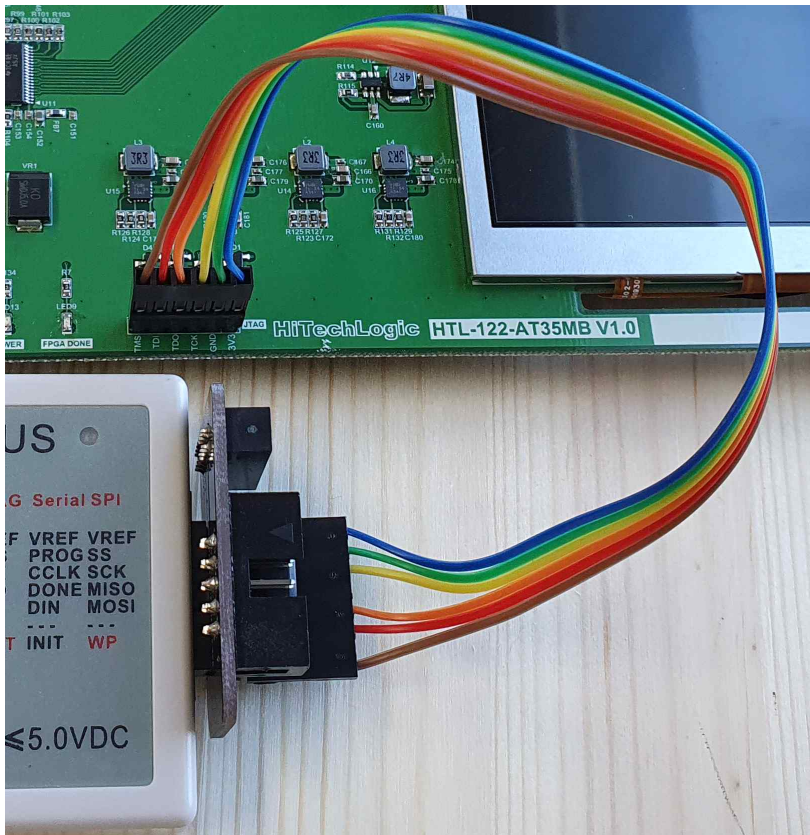


7.2 HTL-123-AT35MB 보드에서는 하이텍로직에 판매하는 XFC-XPCU를 기준으로 설명한다.

7.3 HTL-123-AT35MB 보드의 JTAG 단자는 아래 그림과 같이 표시되어 있다.

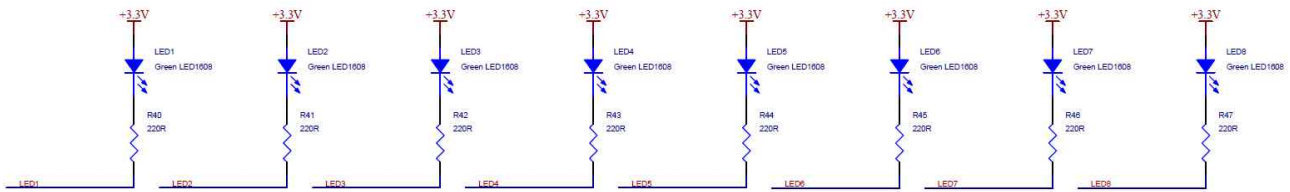


7.4 XFC-XPCU에는 같이 확장보드가 내장되어 있으며, 아래 그림과 같이 케이블을 연결하면 JTAG 연결이 완료되며 프로그램 다운로드가 가능하다.

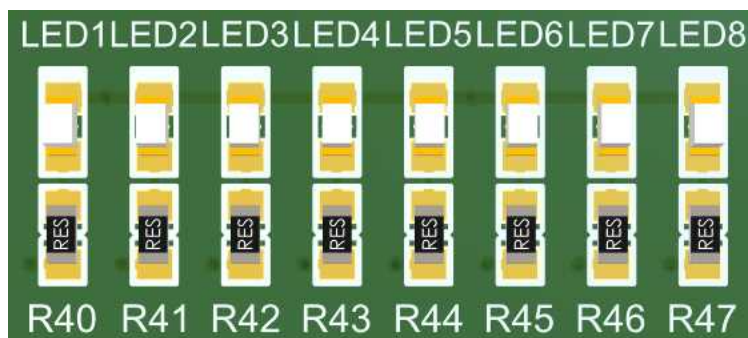


8. User I/O

8.1 8 Output LEDs



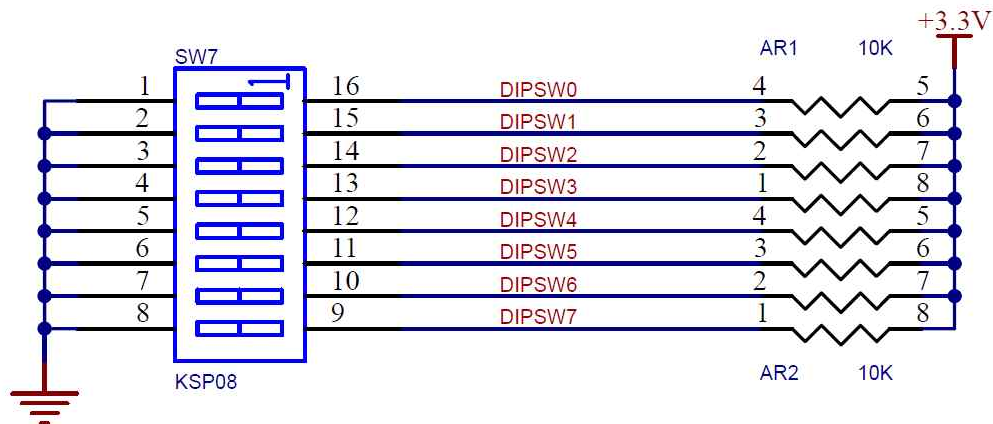
[회로도]



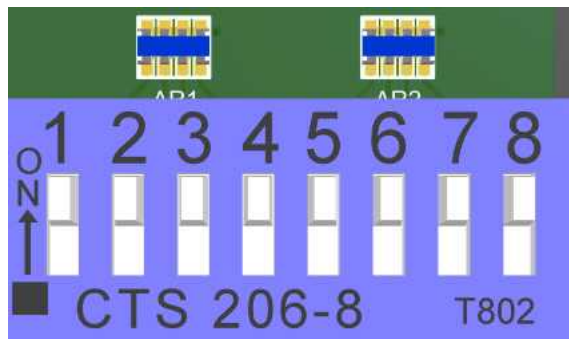
[PCB 3D]

LED	Net Name	FPGA Pin
LED1	LED1	F19
LED2	LED2	D20
LED3	LED3	C20
LED4	LED4	C22
LED5	LED5	B22
LED6	LED6	F21
LED7	LED7	F4
LED8	LED8	G3

8.2 8 Dip Switch



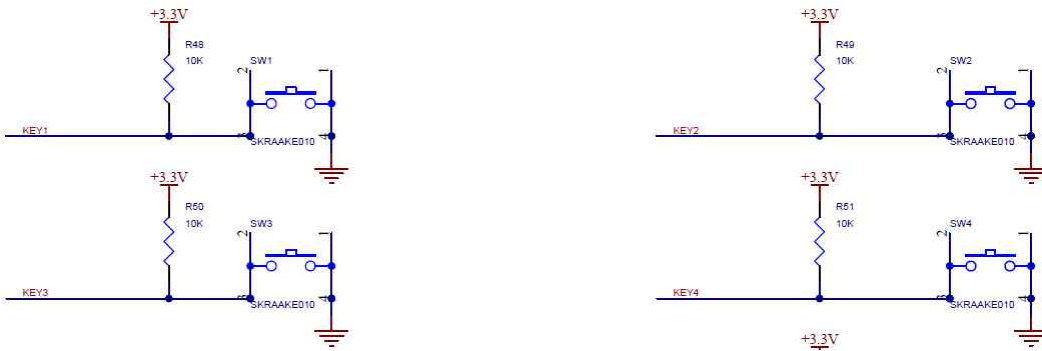
[회로도]



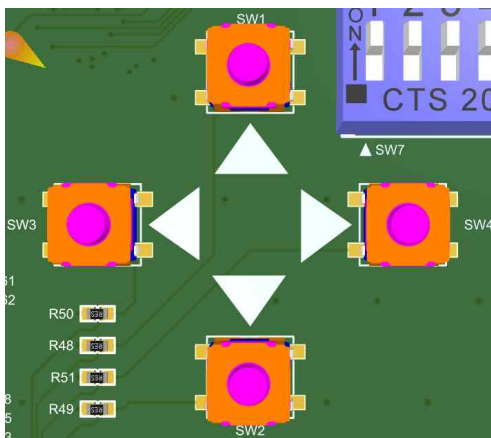
[PCB 3D]

Dip Switch	Net Name	FPGA Pin
SW7	DIPSW0	B1
	DIPSW1	A1
	DIPSW2	C2
	DIPSW3	B2
	DIPSW4	F3
	DIPSW5	E3
	DIPSW6	K2
	DIPSW7	J2

8.3 5 Tact Switch



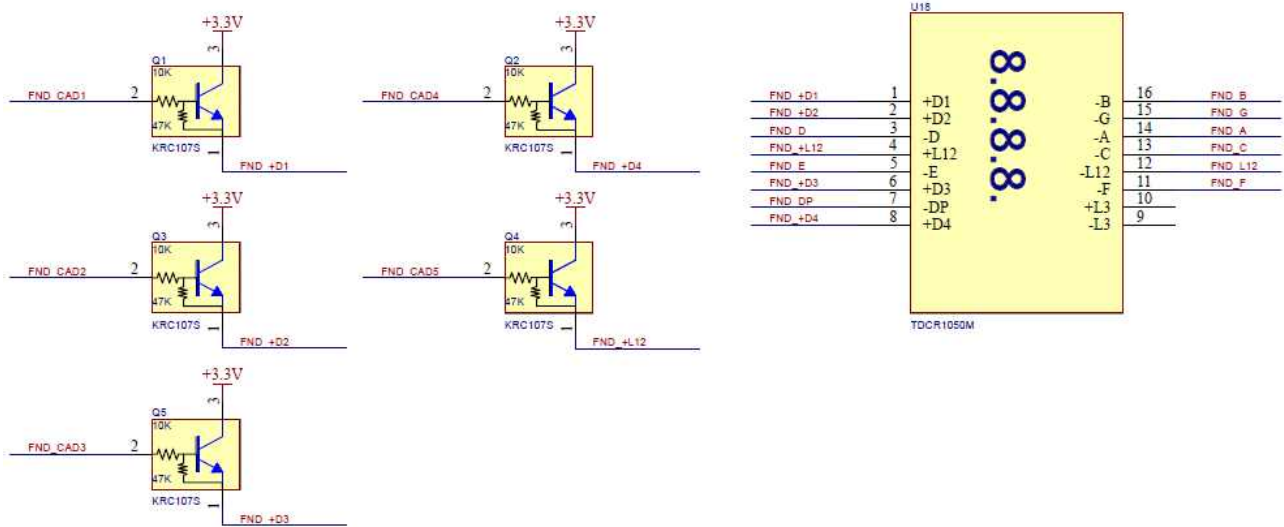
[회로도]



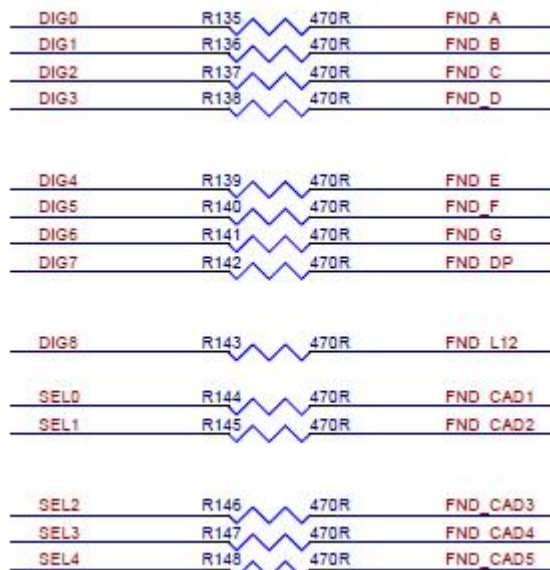
[PCB 3D]

LED	Net Name	FPGA Pin
SW1	KEY1	M13
SW2	KEY2	K14
SW3	KEY3	K13
SW4	KEY4	L13

8.4 Four-Character Seven-Segment



[회로도]



[회로도]



[PCB 3D]

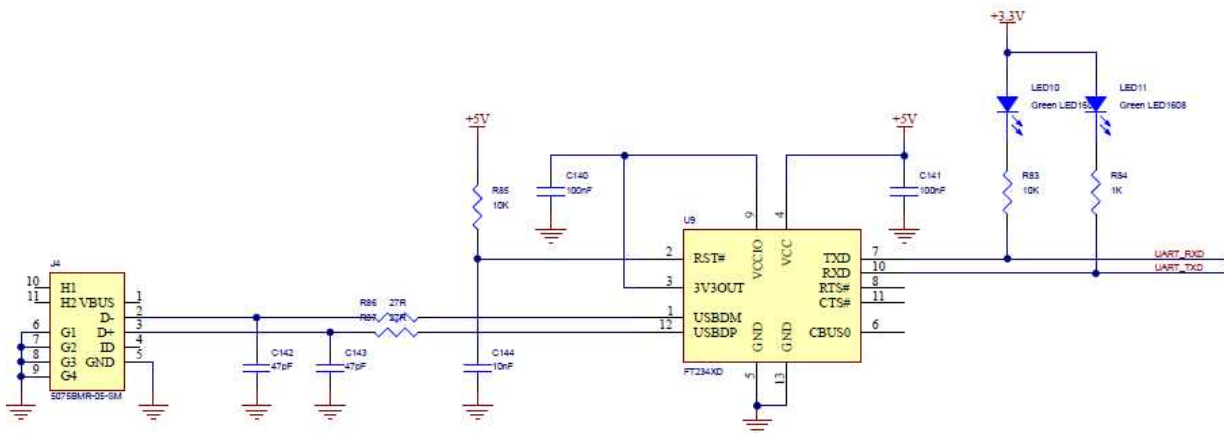
☐ FPGA Connection to Seven-Segment Display

Seven-Segment	Net Name	FPGA Pin
A	DIT0	J5
B	DIT1	M3
C	DIT2	J6
D	DIT3	H5
E	DIT4	G4
F	DIT5	K6
G	DIT6	K3
DP	DIT7	H4

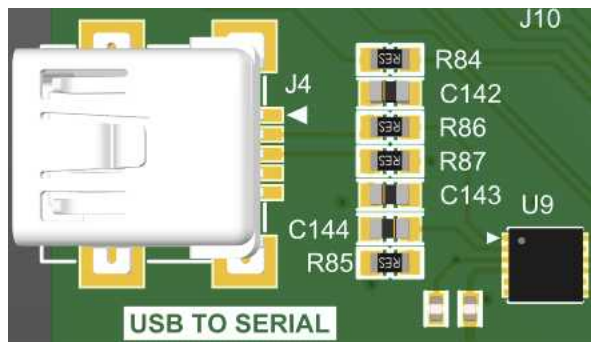
☐ Digit Enable(Anode Control) Signals

Seven-Segment	Net Name	FPGA Pin
SEL0	SEL0	L4
SEL1	SEL1	L5
SEL2	SEL2	
SEL3	SEL3	N4
SEL4	SEL4	M2

8.5 USB-UART Bridge(Serial Port)



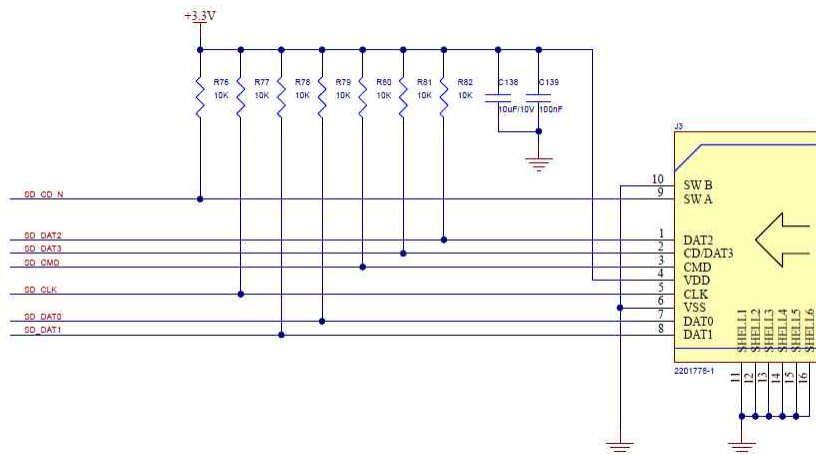
[회로도]



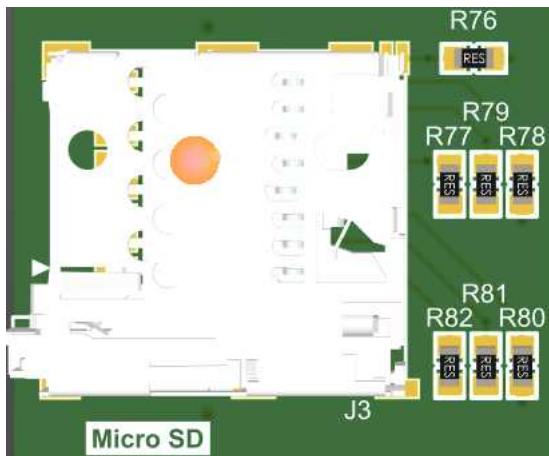
[PCB 3D]

USB to Serial	Net Name	FPGA Pin
TXD	UART_RXD	G15
RXD	UART_RXD	G16

8.6 Mini SD Card



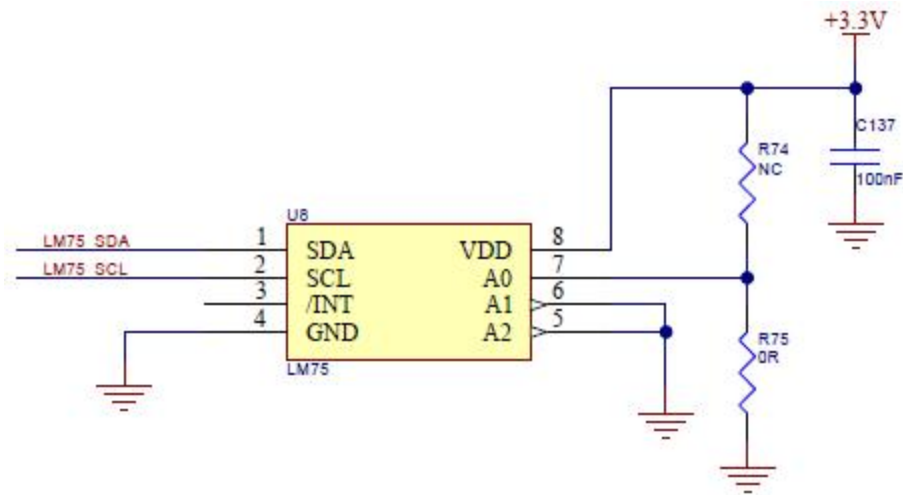
[회로도]



[PCB 3D]

SD Card	Net Name	FPGA Pin
2(CS)	SD_DAT3	N13
3(CMD)	SD_CMD	P15
5(CLK)	SD_CLK	N15
7(DATA)	SD_DAT1	P16

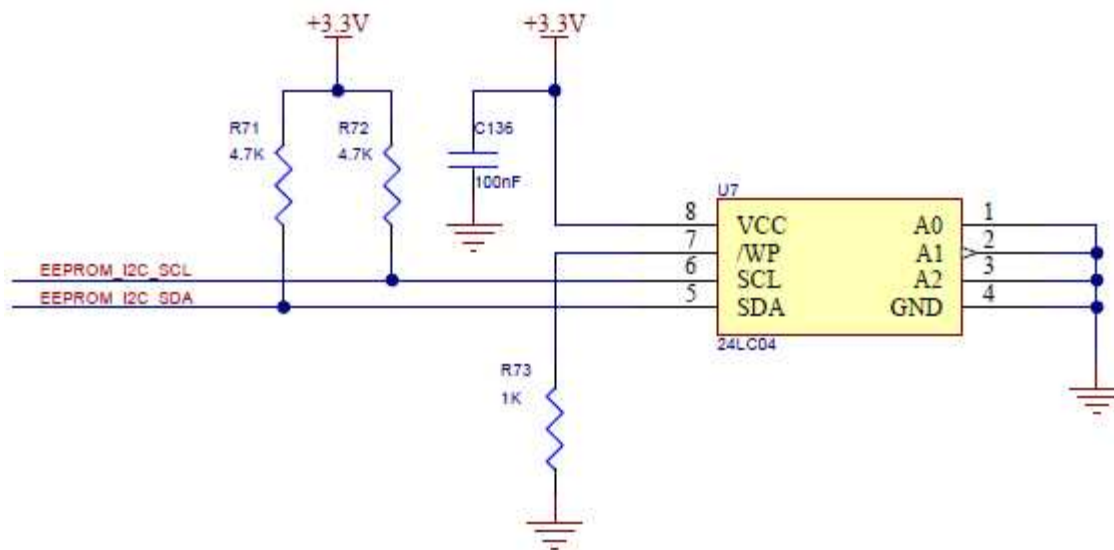
8.7 Temperature Sensor



[회로도]

LM75	Net Name	FPGA Pin
SDA	LM75_SDA	N22
SCL	LM75_SCL	M22

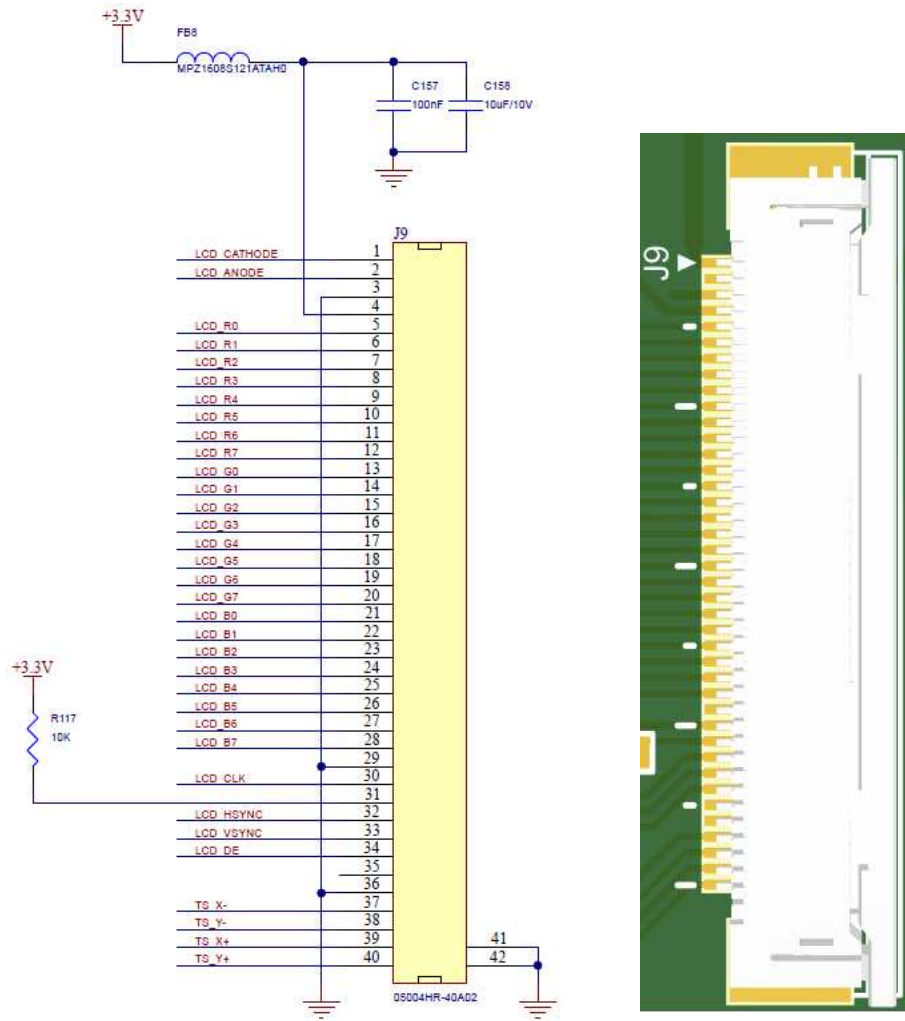
8.8 EEPROM



[회로도]

24LC04	Net Name	FPGA Pin
SDA	EEPROM_I2C_SDA	N19
SCL	EEPROM_I2C_SCL	N18

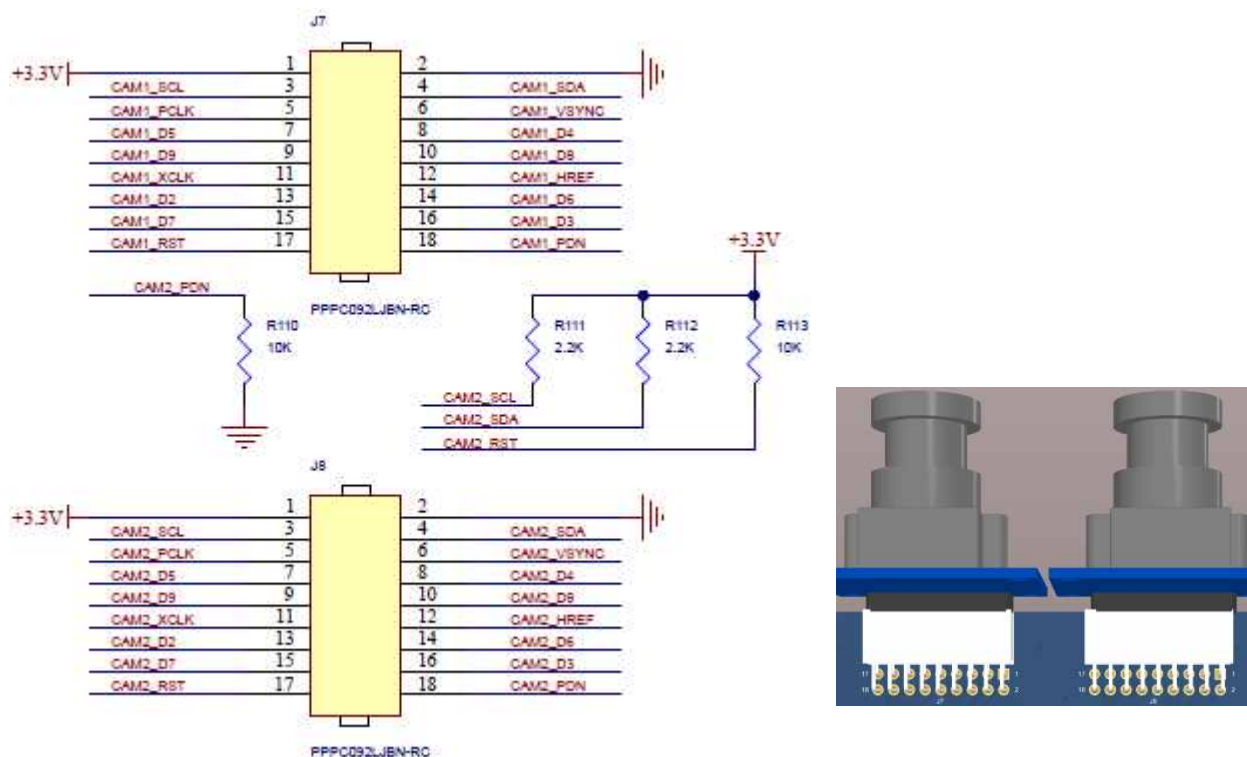
8.9 TFT-LCD



[회로도 및 PCB 3D]

TFT-LCD	Net Name	FPGA Pin
RED0	LCD_R0	D16
RED1	LCD_R1	E16
RED2	LCD_R2	F14
RED3	LCD_R3	F13
RED4	LCD_R4	E14
RED5	LCD_R5	E13
RED6	LCD_R6	D15
RED7	LCD_R7	D14
GREEN0	LCD_G0	B13
GREEN1	LCD_G1	C13
GREEN2	LCD_G2	A14
GREEN3	LCD_G3	A13
GREEN4	LCD_G4	C15
GREEN5	LCD_G5	C14
GREEN6	LCD_G6	A16
GREEN7	LCD_G7	A15
BLUE0	LCD_B0	B16
BLUE1	LCD_B1	B15
BLUE2	LCD_B2	B18
BLUE3	LCD_B3	B17
BLUE4	LCD_B4	A19
BLUE5	LCD_B5	A18
BLUE6	LCD_B6	C19
BLUE7	LCD_B7	C18
PCLK	LCD_CLK	A20
HSYNC	LCD_HSYNC	B20
VSYNC	LCD_VSYNC	C17
DE	LCD_DE	D17

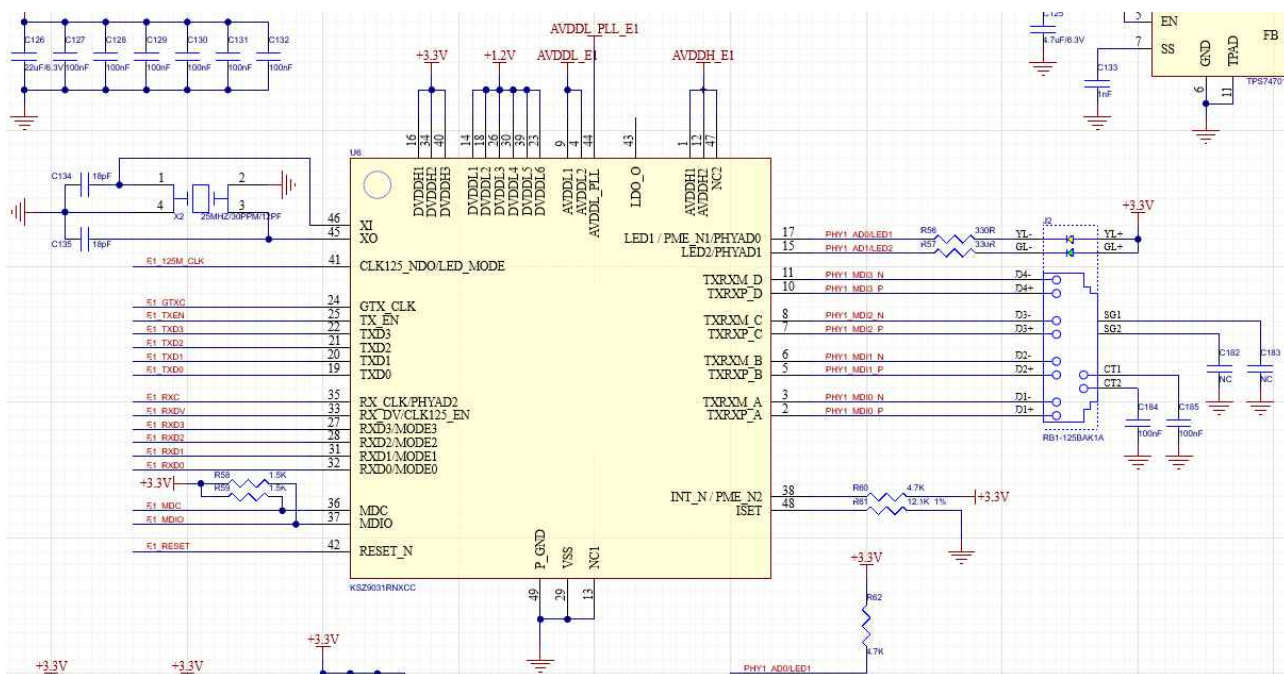
8.10 Dual CMOS Image Sensor(OV5640)



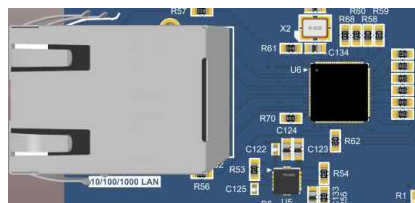
[회로도 및 PCB 3D]

J7	Net Name	FPGA Pin	J8	Net Name	FPGA Pin
SCL	CAM1_SCL	P17	SCL	CAM2_SCL	AA21
SDA	CAM1_SDA	N17	SDA	CAM2_SDA	AA20
PCLK	CAM1_PCLK	J19	PCLK	CAM2_PCLK	W19
VSYNC	CAM1_VSYNC	P19	VSYNC	CAM2_VSYNC	AA19
D3	CAM1_D5	H19	D3	CAM2_D5	W20
D2	CAM1_D4	R18	D2	CAM2_D4	AB20
D7	CAM1_D9	U21	D7	CAM2_D9	AB22
D6	CAM1_D8	T21	D6	CAM2_D8	AB21
XCLK	CAM1_XCLK	V22	XCLK	CAM2_XCLK	AB18
HREF	CAM1_HREF	U22	HREF	CAM2_HREF	AA18
D0	CAM1_D2	V20	D0	CAM2_D2	V19
D4	CAM1_D6	U20	D4	CAM2_D6	V18
D5	CAM1_D7	W22	D5	CAM2_D7	W17
D1	CAM1_D3	W21	D1	CAM2_D3	V17

8.11 Ethernet



[회로도]

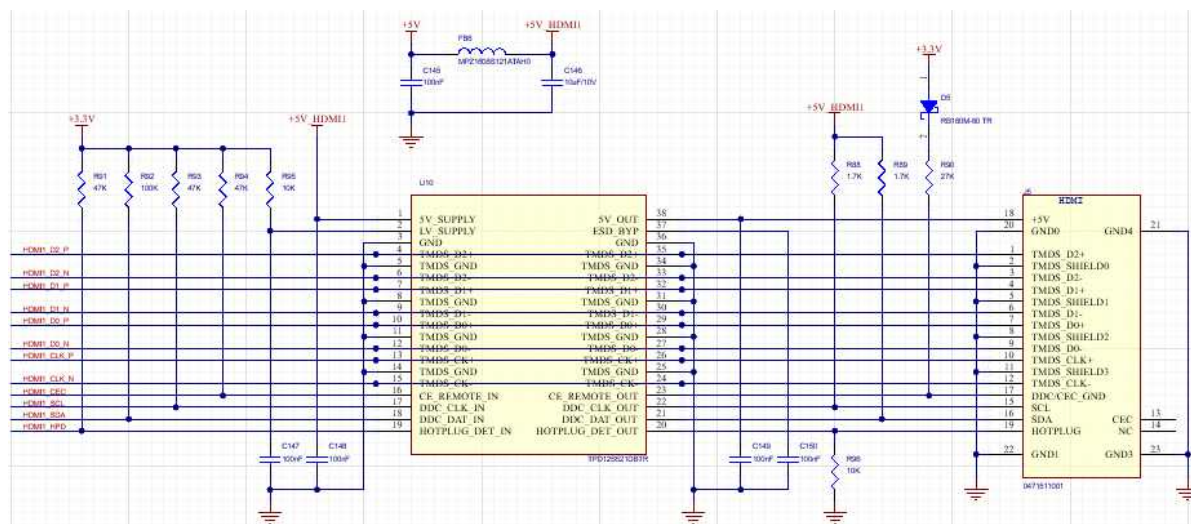


[3D PCB]

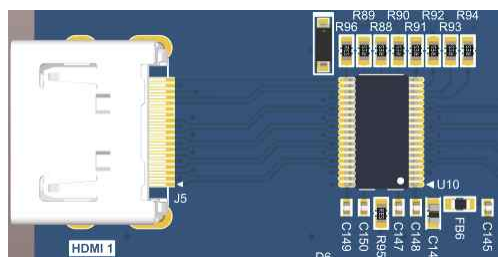
Ethernet	Net Name	FPGA Pin
TXD0	E1_TXD0	J21
TXD1	E1_TXD1	M20
TXD2	E1_TXD2	L18
TXD3	E1_TXD3	L20
RXD0	E1_RXD0	K19
RXD1	E1_RXD1	M15
RXD2	E1_RXD2	J17
RXD3	E1_RXD3	J20
GTX_CLK	E1_GTXC	L14
EX_EN	E1_TXEN	L19
RX_CLK	E1_RXC	K18
RX_DV	E1_RXDV	M21
MDC	E1_MDC	K17
MDIO	E1_MDIO	K16
RESET_N	E1_RESET	L15

8.12 HDMI

- HDMI1 : HDMI Output



[회로도]



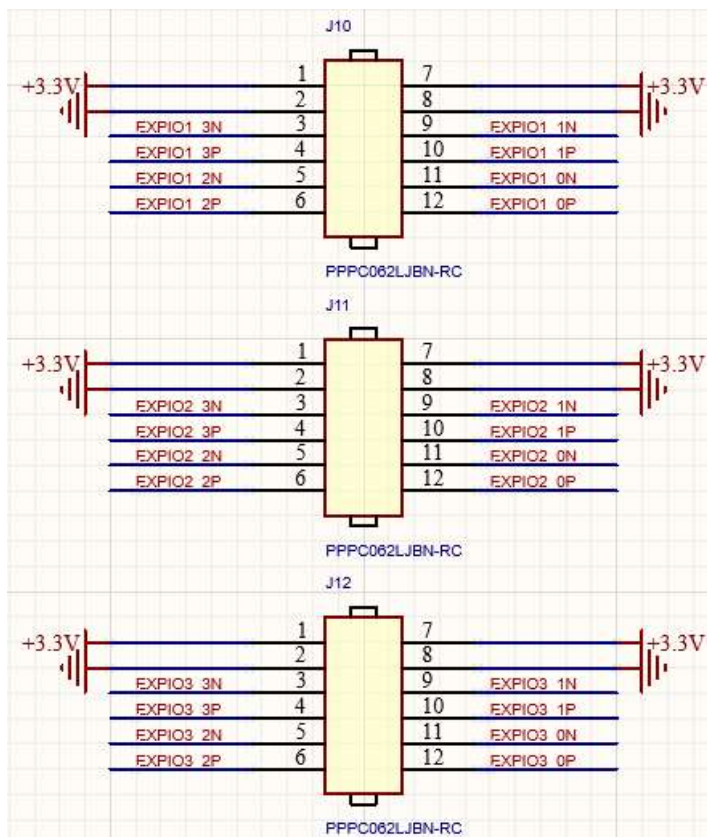
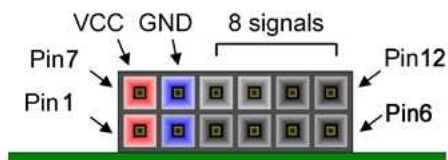
[3D PCB]

Ethernet	Net Name	FPGA Pin
HDMI1_CLK_P	HDMI1_CLK_P	E1
HDMI1_CLK_N	HDMI1_CLK_N	D1
HDMI1_D0_P	HDMI1_D0_P	G1
HDMI1_D0_N	HDMI1_D0_N	F1
HDMI1_D1_P	HDMI1_D1_P	H2
HDMI1_D1_N	HDMI1_D1_N	G2
HDMI1_D2_P	HDMI1_D2_P	K1
HDMI1_D2_N	HDMI1_D2_N	J1
HDMI1_SCL	HDMI1_SCL	P4
HDMI1_SDA	HDMI1_SDA	N3
HDMI1_OUT_EN	HDMI1_OUT_EN	M6
HDMI1_HPD	HDMI1_HPD	P5

[illegible]

Ethernet	Net Name	FPGA Pin
HDMI2_CLK_P	HDMI2_CLK_P	K4
HDMI2_CLK_N	HDMI2_CLK_N	J4
HDMI2_D0_P	HDMI2_D0_P	M1
HDMI2_D0_N	HDMI2_D0_N	L1
HDMI2_D1_P	HDMI2_D1_P	P2
HDMI2_D1_N	HDMI2_D1_N	N2
HDMI2_D2_P	HDMI2_D2_P	R1
HDMI2_D2_N	HDMI2_D2_N	P1
HDM2_SCL	HDMI2_SCL	N5
HDMI2_SDA	HDMI2_SDA	L6
HDMI2_OUT_EN	HDMI2_OUT_EN	P6
HDMI2_HPD	HDMI2_HPD	M5

9. Expansion Connectors



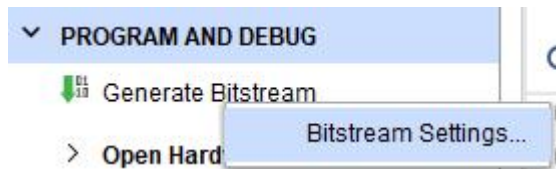
J10	Net Name	FPGA	J11	Net Name	FPGA	J12	Net Name	FPGA
1	3V3		1	3V3		1	3V3	
2	GND		2	GND		2	GND	
3	EXPIO1_3N	H18	3	EXPIO2_3N	B21	3	EXPIO3_3N	D2
4	EXPIO1_3P	H17	4	EXPIO2_3P	A21	4	EXPIO3_3P	E2
5	EXPIO1_2N	H15	5	EXPIO2_2N	K22	5	EXPIO3_2N	G22
6	EXPIO1_2P	J15	6	EXPIO2_2P	K21	6	EXPIO3_2P	G21
7	3V3		7	3V3		7	3V3	
8	GND		8	GND		8	GND	
9	EXPIO1_1N	G18	9	EXPIO2_1N	G20	9	EXPIO3_1N	D21
10	EXPIO1_1P	G17	10	EXPIO2_1P	H20	10	EXPIO3_1P	E21
11	EXPIO1_0N	H14	11	EXPIO2_0N	H22	11	EXPIO3_0N	D22
12	EXPIO1_0P	J14	12	EXPIO2_0P	J22	12	EXPIO3_0P	E22

10. 예제 목록

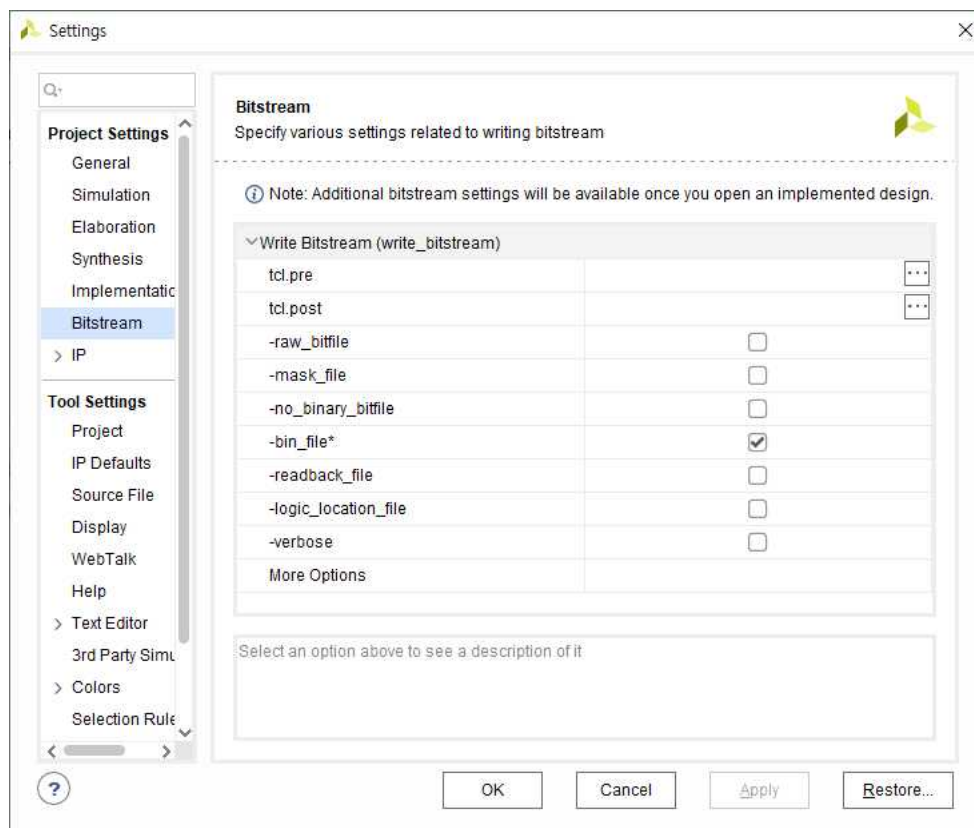
명칭	VHDL	Verilog
I/O	p01_led_on	p01_led_on
	p02_led_cnt	p02_led_cnt
	p03_led_dipsw	p03_led_dipsw
	p04_Seven_Segment	p04_Seven_Segment
	p05_uart	p05_uart
		p06_eeprom
TFT-LCD	p01_color_fill_red	p01_vertical_pattern
	p02_color_pattern	p02_horizontal_pattern
	p03_font_8_16_position	
	p04_font_8_16_num_display	
	p05_font_8_16_alphabet_display	
	p06_font_24_24_display	
SD Card		p01_sd_picture_tftlcd430
		p02_sd_picture_hdmi
HDMI & TFT-LCD & OV5640		p01_hdmi_output_test
		p02_hdmi_loop
		p03_dual_ov5640_ddr3_tftlcd430
		p04_dual_ov5640_ddr3_hdmi
Ethernet & OV5640		p01_ethernet_test
		p02_ethernet_to_video

11. FPGA Configuration

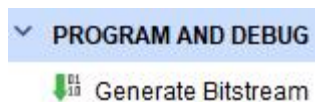
- Vivado를 실행하고 Generate Bitstream의 팝업 메뉴인 Bitstream Settings를 클릭한다.



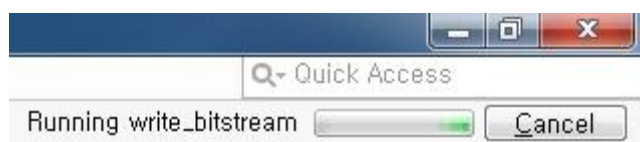
- bin_file 우측의 체크 박스를 체크한다.



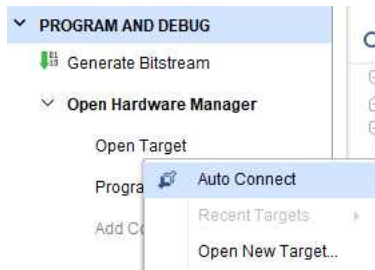
- Generate Bitstream을 클릭한다.



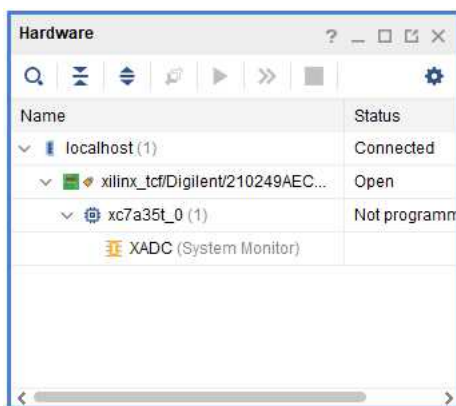
- Bitstream을 생성 중에는 우측 상단에 아래 그림과 같이 표시된다.



- HTL-122-AT35MB 보드에 JTAG을 연결한 후 전원을 인가한다.
- Open Hardware Manager → Open Target → Auto Connect를 클릭한다.



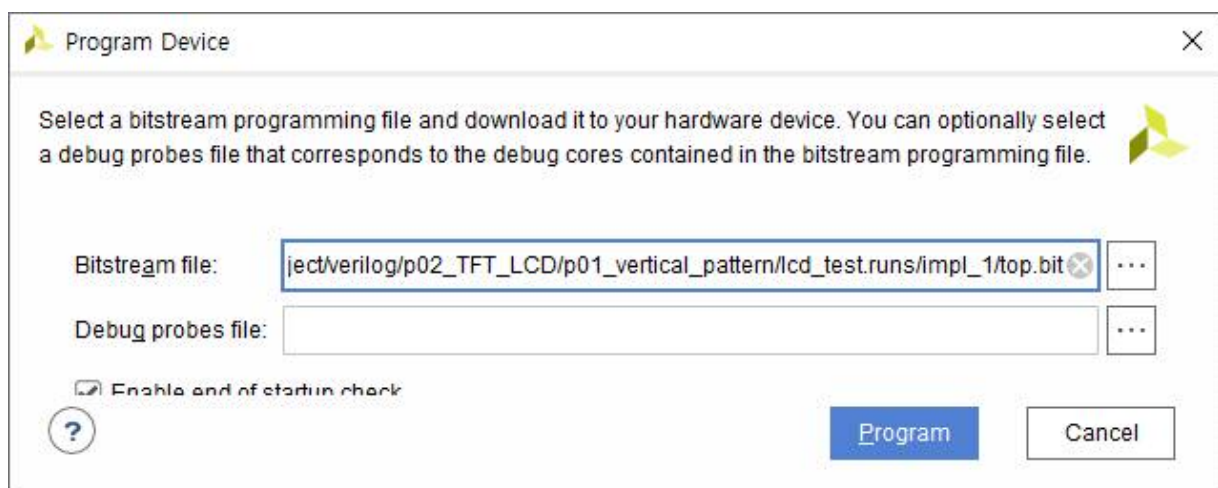
- 하드웨어 적인 문제가 발생하지 않았다면 아래 그림과 같이 표시된다.



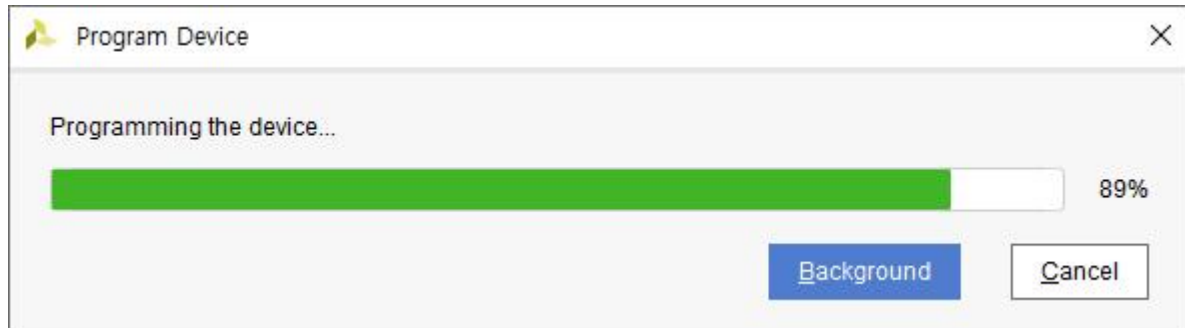
- Program Device 팝업 메뉴 중에서 xc7a35t_0를 클릭한다.



- 현재 작업 중인 프로젝트의 bit 파일을 선택하고 Progr... 버튼을 클릭한다.



- 아래 그림과 같이 진행률이 표시된다.

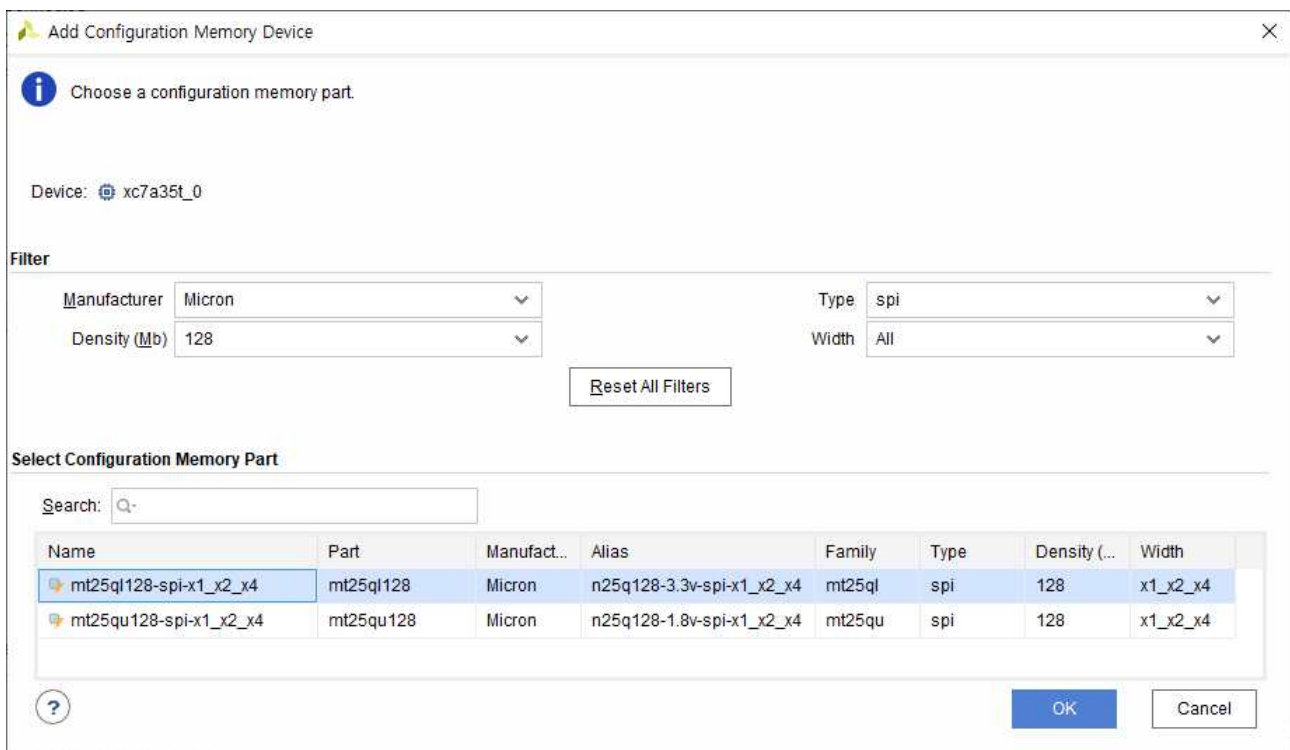


12. Configuration Device Programming

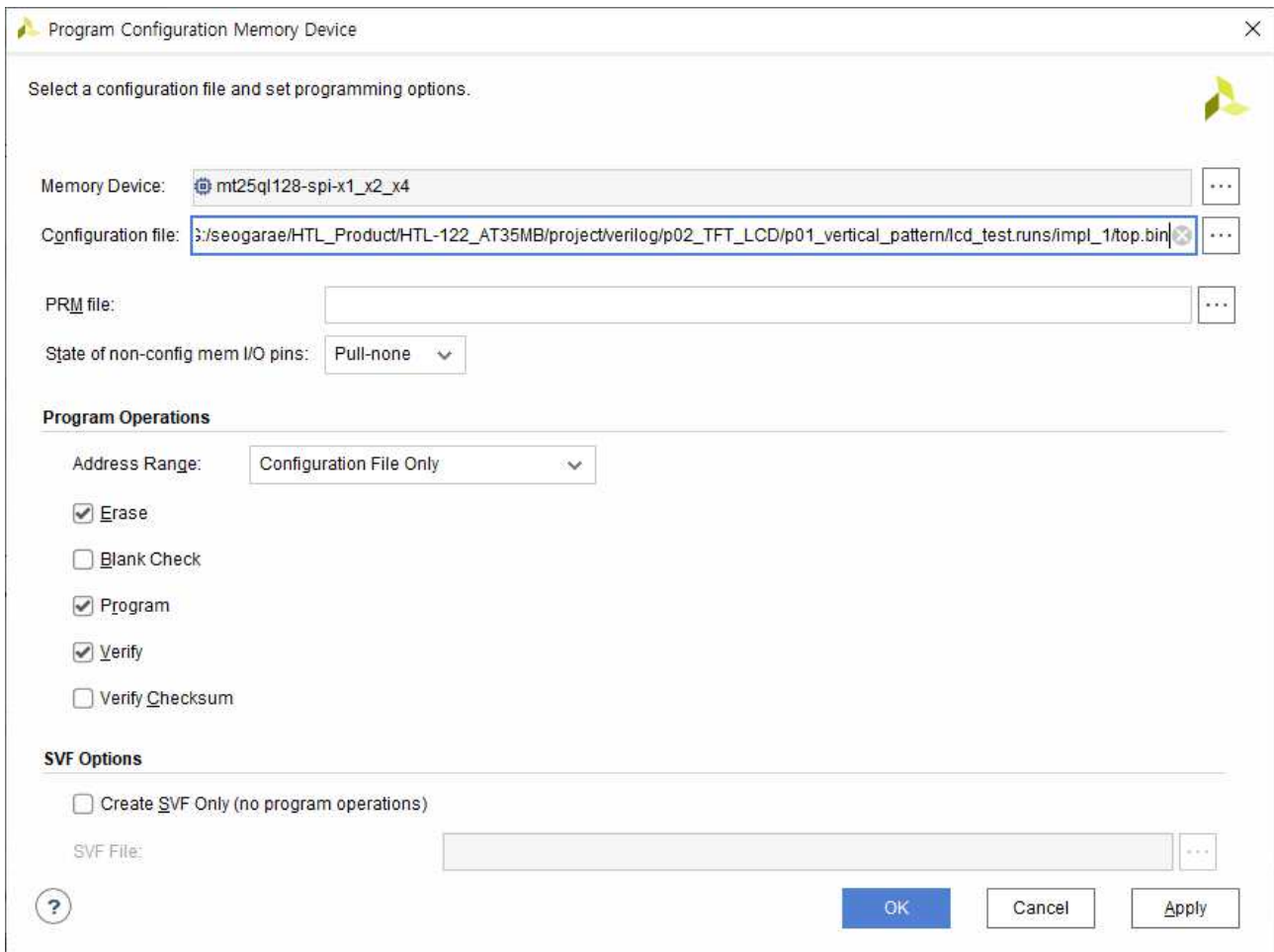
- Add Configuration Memory 업 메뉴 중에서 xc7a35t_0를 클릭한다.



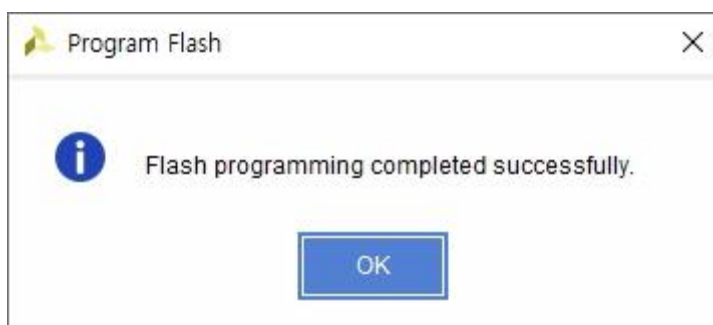
- 아래 그림과 같이 디바이스(n25q128-3.3...)를 선택한다.



- Configuration file의 브라우저 버튼을 클릭하여, 현재 작업중인 프로젝트의 bin 파일을 선택하고, OK 버튼을 클릭한다.



- 완료되면 아래 그림과 같은 메시지가 출력된다.



13. 회로도(제품구매 시 제공)